MASTER WORKFLOW

1. Find $\mathbf{B}$ from source
2. Compute $\partial\mathbf{B}/\partial t$ (or Φ then $d\Phi/dt$)
3. Set up $d\mathbf{s}$ from geometry
4. Dot product: $\partial\mathbf{B}/\partial t \cdot d\mathbf{s}$
5. Integrate over surface area
6. Apply negative sign $\Rightarrow$ EMF

EMF EQUATIONS

Faraday's Law: $V_{\text{emf}} = -\dfrac{d\Phi}{dt}$

Transformer EMF (loop stationary): $V_{\text{emf}}^{\text{tr}} = -\displaystyle\int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$

Motional EMF (loop moving): $V_{\text{emf}}^{\text{mot}} = \displaystyle\oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$

Total EMF: $V_{\text{emf}} = V_{\text{emf}}^{\text{tr}} + V_{\text{emf}}^{\text{mot}}$

Generator (rotating loop): $V_{\text{emf}} = A\omega B_0 \sin(\omega t + C_0)$

Multi-turn coil (N turns): $V_{\text{emf}} = -N\dfrac{d\Phi}{dt}$

MAGNETIC FLUX

General: $\Phi = \displaystyle\int_S \mathbf{B} \cdot d\mathbf{s}$

Uniform B, area A: $\Phi = B \cdot A$

MAGNETIC FIELD SOURCES

Long straight wire: $\mathbf{B} = \hat{\phi}\,\dfrac{\mu_0 I}{2\pi r}$

Wire in medium μ: $\mathbf{B} = \hat{\phi}\,\dfrac{\mu I(t)}{2\pi r}$

Iron core: $\mu = \mu_r \mu_0$

SURFACE ELEMENTS $d\mathbf{s}$

Loop geometry	$d\mathbf{s}$
Flat in xy-plane	$\hat{z}\,ds$
Normal to $\hat{\phi}$ (toroid)	$\hat{\phi}\,dr\,dz$
Normal to $\hat{r}$	$\hat{r}\,d\phi\,dz$
Normal to $\hat{z}$ (cylinder)	$\hat{z}\,r\,dr\,d\phi$

DOT PRODUCT RULES

$\hat{x}\cdot\hat{x} = \hat{y}\cdot\hat{y} = \hat{z}\cdot\hat{z} = \hat{r}\cdot\hat{r} = \hat{\phi}\cdot\hat{\phi} = 1$

Any two *different* unit vectors $= 0$

e.g. $(\hat{\phi}\,B_0)\cdot(\hat{\phi}\,ds) = B_0\,ds$

KEY DERIVATIVES & INTEGRALS

$\dfrac{d}{dt}\sin(\omega t) = \omega\cos(\omega t)$ $\dfrac{d}{dt}\cos(\omega t) = -\omega\sin(\omega t)$

$\displaystyle\int_a^b \frac{dr}{r} = \ln\left(\frac{b}{a}\right)$

$\displaystyle\int_S ds = A$ (uniform B, pull out of integral)

$\displaystyle\int_{z_0}^{z_0+c} dz = c$

CURRENT DIRECTION RULE

$V_{\text{emf}} < 0 \Rightarrow I$ flows $-$ to $+$ inside source $\Rightarrow$ direction is $-\hat{\phi}$

$V_{\text{emf}} > 0 \Rightarrow I$ flows $+$ to $-$ inside source $\Rightarrow$ direction is $+\hat{\phi}$

$V_{\text{emf}} = 0 \Rightarrow$ no current

LENZ'S LAW (INTUITION)

Induced I opposes the *change* in flux.

$\Phi \uparrow$ (B into page increasing)

$\Rightarrow$ induced I creates B out of page.

CIRCUIT WITH EMF SOURCE

Current: $I = \dfrac{V_{\text{emf}}}{R + R_i}$

R = external resistance, R_i = loop internal resistance

DISPLACEMENT & CONDUCTION CURRENT

Conduction current density: $\mathbf{J}_c = \sigma\mathbf{E}$

Displacement current density: $\mathbf{J}_d = \epsilon\dfrac{\partial\mathbf{E}}{\partial t}$

Displacement current (capacitor): $I_d = \dfrac{\epsilon_r\epsilon_0 A}{d}\dfrac{\partial V}{\partial t}$

Conduction current (capacitor): $I_c = \dfrac{\sigma V A}{d}$

Parallel-plate E field: $E = V/d$

LOSSY CAPACITOR EQUIV. CIRCUIT

Resistance: $R = \dfrac{d}{\sigma A}$

Capacitance: $C = \dfrac{\epsilon_r\epsilon_0 A}{d}$

$R \| C$ in parallel. $I_c \propto V \Rightarrow$ resistor; $I_d \propto \partial V/\partial t \Rightarrow$ capacitor.

CHARGE RELAXATION

Charge decay: $\rho_v(t) = \rho_{v0}\,e^{-t/\tau_r}$

Relaxation time constant: $\tau_r = \dfrac{\epsilon}{\sigma} = \dfrac{\epsilon_r\epsilon_0}{\sigma}$

Larger $\tau_r \Rightarrow$ better insulator

PHASOR BASICS

Definition: $f(t) = \mathrm{Re}\{\tilde{F} e^{j\omega t}\}$

Time deriv. in phasor domain: $\partial/\partial t \rightarrow j\omega$

Find $\tilde{\mathbf{H}}$ from $\tilde{\mathbf{E}}$: $\tilde{\mathbf{H}} = -\dfrac{1}{j\omega\mu}\nabla \times \tilde{\mathbf{E}}$

Faraday (phasor): $\nabla \times \tilde{\mathbf{E}} = -j\omega\mu\tilde{\mathbf{H}}$

Ampere (phasor): $\nabla \times \tilde{\mathbf{H}} = (j\omega\epsilon + \sigma)\tilde{\mathbf{E}}$

PHASOR TRICK: j AND $-j$

$\mathrm{Re}\{e^{j\omega t}\} = \cos(\omega t)$

$\mathrm{Re}\{je^{j\omega t}\} = -\sin(\omega t)$

$\mathrm{Re}\{-je^{j\omega t}\} = \sin(\omega t)$

$\sin(\omega t) = \cos(\omega t - \pi/2)$

So: $3\sin(\omega t - kz) = \mathrm{Re}\{-3j\, e^{j(\omega t - kz)}\}$

CURL IN CARTESIAN

$$\nabla \times \mathbf{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ A_x & A_y & A_z \end{vmatrix}$$

$\hat{x}$ comp: $\partial_y A_z - \partial_z A_y$

$\hat{y}$ comp: $\partial_z A_x - \partial_x A_z$

$\hat{z}$ comp: $\partial_x A_y - \partial_y A_x$

WAVE PROPAGATION

$+z$ **direction:** $f = F\cos(\omega t - kz)$

$-z$ **direction:** $f = F\cos(\omega t + kz)$

Phase velocity: $u_p = \omega/k = 1/\sqrt{\mu\epsilon}$

Intrinsic impedance: $\eta = \sqrt{\mu/\epsilon}$

Free space: $\eta_0 = 377\ \Omega,\ c = 3 \times 10^8$ m/s

CONSTANTS

Constant	Value
μ_0	$4\pi \times 10^{-7}$ H/m
ϵ_0	8.854×10^{-12} F/m
ϵ_0 approx	$\frac{1}{36\pi} \times 10^{-9}$ F/m
c	3×10^8 m/s
η_0	$377\ \Omega$

UNIT CONVERSIONS

Expression	Unit
Tm^2/s	V (volt)
H·A/s	V (volt)
V/(Ω)	A (ampere)
F/m · V/s	$\mathrm{A/m}^2$
σ: S/m	ϵ: F/m, μ: H/m

FREQUENCY CONVERSIONS

$\omega = 2\pi f$ (rad/s from Hz)

$f = \omega/(2\pi)$

RPM → rad/s: $\omega = 2\pi \cdot \dfrac{\mathrm{RPM}}{60}$

CYLINDRICAL COORDINATES

(r, ϕ, z): $x = r\cos\phi,\ y = r\sin\phi$

$dV = r\,dr\,d\phi\,dz$

Cross products:

$\hat{r} \times \hat{\phi} = \hat{z},\ \hat{\phi} \times \hat{z} = \hat{r},\ \hat{z} \times \hat{r} = \hat{\phi}$

PROBLEM TYPE LOOKUP

Scenario	Key formula
Stationary loop, varying B	Transformer EMF
Moving loop, static B	Motional EMF
Long wire + loop	$B = \hat{\phi}\mu I/2\pi r$, integrate ln
Toroid, N turns	$V = -N\,d\Phi/dt$
Rotating loop	$V = A\omega B_0 \sin\omega t$
Lossy capacitor	$R\|\|C$ parallel circuit
Charge relaxation	$\rho_v = \rho_{v0} e^{-t/\tau_r}$
H from **E**	Phasor $\nabla \times \tilde{\mathbf{E}}$